House Price Prediction – Summary Report

Objective

The objective of this project was to analyze housing data and develop machine learning models capable of predicting house prices based on property characteristics such as area, number of bedrooms, bathrooms, parking availability, and additional amenities.

Data Preparation

The dataset was cleaned by removing duplicate records, handling missing values, and converting categorical variables into numerical form using one-hot encoding. This ensured that the machine learning algorithms could effectively process all available information.

Model Performance

Two regression models were developed and evaluated:

- Linear Regression
    - MAE
    - RMSE
    - R^2 Score
- Random Forest Regressor

The Random Forest model achieved the better predictive performance, indicating that house prices are influenced by non-linear relationships among features.

Key Findings

- House area showed a strong positive relationship with price.
- The number of bathrooms and parking spaces significantly increased property value.
- Houses located in preferred areas generally commanded higher prices.
- Additional amenities such as air conditioning and furnished status contributed to higher selling prices.

Business Recommendation

Real estate businesses should prioritize marketing properties with larger living areas, multiple bathrooms, parking facilities, and premium amenities. These features consistently contribute to higher market values and stronger buyer demand.

Conclusion

The project successfully demonstrated how machine learning can be used to estimate housing prices. The developed models provide valuable insights for property valuation and can assist buyers, sellers, and real estate agencies in making data-driven decisions.